Dynamical Localization for General Scattering Quantum Walks

QuantAlps Days 2026

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Motivation

Key messages

- Quantum Walks (QWs) as mathematical objects are interesting to study, but also have numerous applications in quantum computing and physics
- Low level of randomness already leads to localization

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QWs are quantum mechanical analogue of classical Random Walks
They model a quantum particle moving on a graph in **discrete time**
Described by unitary operator U acting on Hilbert space

$$i\frac{d}{dt}\psi(t) = H\psi(t) \longrightarrow \psi(t) = \big(\underbrace{e^{-iH}}_{U}\big)^t\psi_0$$

Discrete time, e.g. look at $t = 0, 1, 2, ...$

Applications

Quantum Computing

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- use *ballistic* transport
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Physics

- Simple models of quantum transport
- condensed matter physics: introduce disorder / boundaries, then study Anderson localization, topological edge states, bandstructure
- experimentally realizable: photons, trapped ions, superconducting circuits

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Local dynamics at $x \in V$ governed by scattering matrix $S(x) \in \mathcal{U}(d_x)$, that maps incoming onto outgoing edges

$$U\,|xy\rangle = \sum_{z \sim x} S_{zy}(x)\,|zx\rangle\,.$$

Matrix elements of $S(x)$ are labeled by neighbors y, z of x

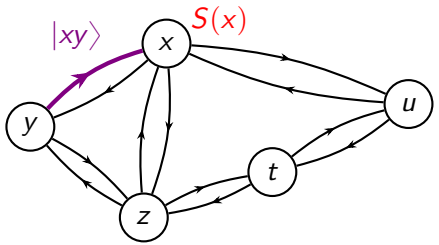
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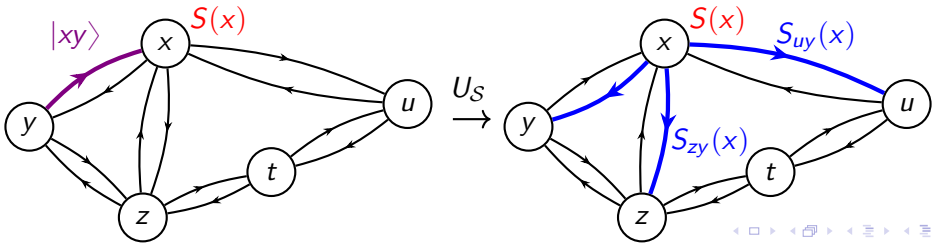
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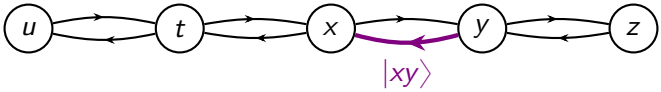
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Scattering Quantum Walk on a line

Vertices $V = \mathbb{Z}$, edges D between neighbors
Scattering matrices $S(x)$ are 2×2 **unitary** matrices

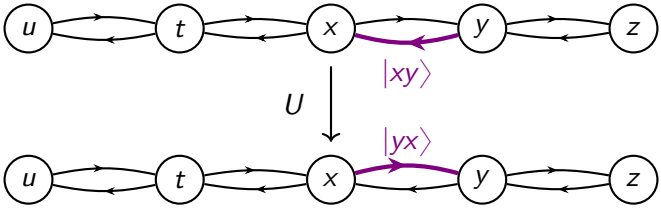
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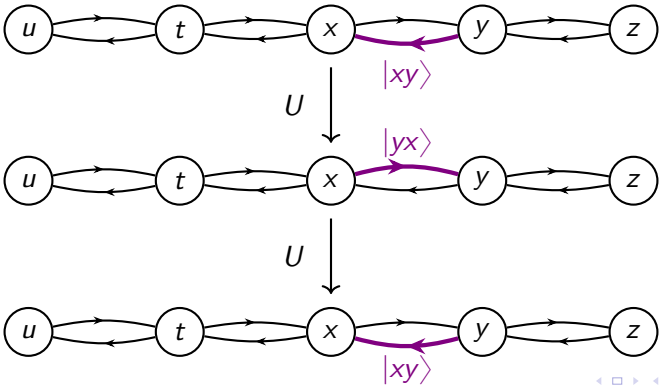
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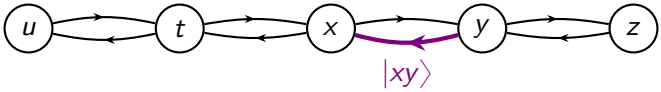
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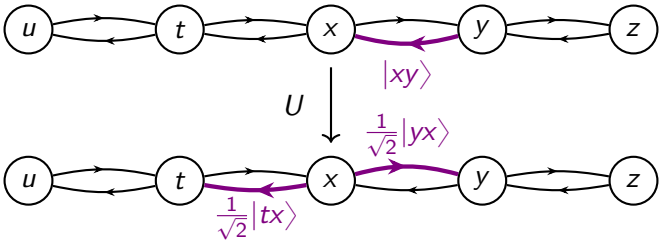
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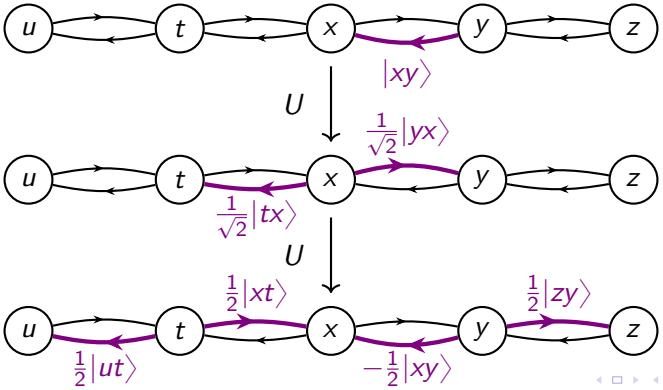
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Difference between classical RW and QW

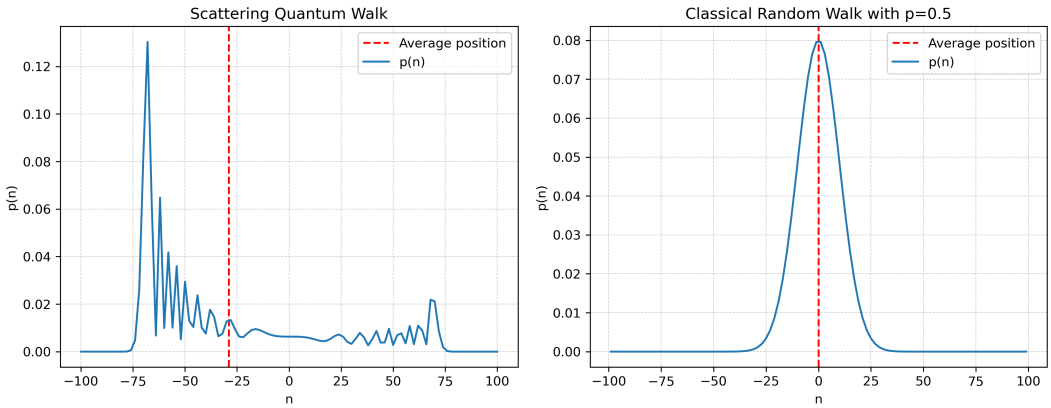


Figure 4: Comparison of a Quantum and a classical Random Walk after $t = 100$ steps.

Differences: limiting distribution, transport, speed of "spreading" ($\sigma = t$ vs $\sigma = \sqrt{t}$)

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Equip every vertex x with random variable $\omega_x \in [0, 2\pi]$, define random scattering matrix:

$$S_\omega(x) = e^{i\omega_x} S(x).$$

Defines random QW U_ω
Intuition: Randomness smooths out the perturbation

Dynamical Localization

Theorem (Joye, S., Warzel '26 (hopefully))

Let G have bounded degree and ω_x be i.i.d. with bounded density, then $\forall\, g > 0\ \exists\, \varepsilon, c > 0$ such that if $\sup_x \|S(x) - I\| \leqslant \varepsilon$, then $\forall\ |xy\rangle, |x'y'\rangle \in \ell^2(D)$

$$\mathbb{E}\Big(\sup_{n\in\mathbb{Z}} |\langle xy \,|\, U_\omega^n\, x'y'\rangle| \Big) \leqslant c\, e^{-g\, dist(x,x')}.$$

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Interpretation: Probability to move from x to x' decreases on average exponentially in the distance between x and x', independently of how many steps the QW may take

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Fractional Moment Criterion

Usually: Fractional Moment Estimates $\Longrightarrow$ Dynamical Localization
for some $s \in (0, \frac{1}{3})$ and all scattering matrices s.t. $\sup_{x \in V} \|S(x) - I\| \leqslant \varepsilon$:

$$\sup_{|z| \neq 1} \mathbb{E}\big(|\langle xy\,|\,(U_\omega - z)^{-1}\, x'y'\rangle|^s\big) \leqslant c\, e^{-g\mathrm{dist}(x,x')}$$

Why is it not much randomness?

Isolate randomness: $U_\omega = D_\omega U$ with $D_\omega |xy\rangle = e^{i\omega_y} |xy\rangle$, write D_ω as matrix:

$$D_\omega = \begin{pmatrix} \ddots & & & & & & \\ & e^{i\omega_x} & & & & & \\ & & e^{i\omega_x} & & & & \\ & & & e^{i\omega_x} & & & \\ & & & & e^{i\omega_y} & & \\ & & & & & e^{i\omega_y} & \\ & & & & & & e^{i\omega_y} \\ & & & & & & & \ddots \end{pmatrix}$$

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However: Fractional moment criterion requires independent phases
$\longrightarrow$ Use Eigenfunction Correlators to bridge the gap

The effects of Randomness

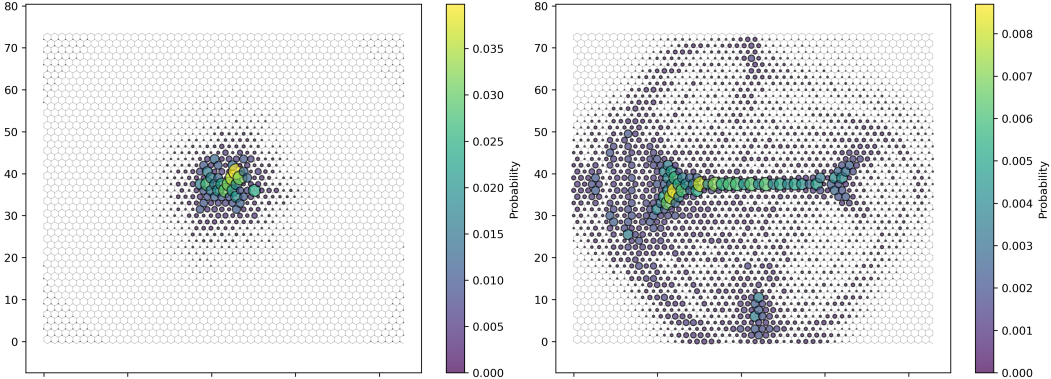


Figure 5: Simulation of a Quantum Walk on the hexagonal lattice with and without randomness

Thank you for your attention!